

Joint 2-D Sensor Location Calibration and DOA Estimation for Virtual Coprime Arrays with a Single Movable Antenna

Qixuan Lu, Jingjing Pan, Weiyi Sun, Meng Sun and Yuehe Ning
College of Electronic and Information Engineering
Nanjing University of Aeronautics and Astronautics
Nanjing, China
qixuanlu@nuaa.edu.cn

Abstract—Movable antennas (MAs) offer advantages for Direction of Arrival (DOA) estimation, owing to their low hardware cost and flexible deployment. By mounting antennas on a movable platform, an extended virtual array can be formed. However, the mobility introduces the challenge of sensor location errors. This paper considers the scenarios with 2-D sensor location errors, where a single MA collects data at coprime array positions to form a virtual coprime array. The structure of the coprime array is then further exploited to generate a virtual uniform linear array. A maximum likelihood-based method is proposed to calibrate the sensor positions. Simulation results demonstrate that the proposed algorithm effectively performs position calibration and improves estimation performance.

Index Terms—Direction of arrival (DOA), Movable antenna, Sensor location errors, Coprime array.

I. INTRODUCTION

Direction of arrival (DOA) estimation stands as one of the core technologies in the field of array signal processing, playing a critical role in numerous applications such as radar, sonar, wireless communications and electronic warfare [1], [2]. The early development of DOA estimation primarily focused on fixed antenna arrays, with the uniform linear array (ULA) being the most fundamental and extensively studied configuration. However, the sensor spacing is typically set to half the wavelength in the ULA which induces mutual coupling and subsequently degrades the accuracy of DOA estimation.

To reduce mutual coupling, sparse arrays, such as minimum redundant array [3], coprime array [4], [5], [6], [7] and nested array [8], are proposed. Although the sparse array has less mutual coupling compared to the ULA, its array geometry is difficult to adjust once fixed. To mitigate antenna costs and enable flexible array deployment, movable antennas (MAs) are employed [9]. MAs are typically deployed on movable platforms, such as unmanned aerial vehicles (UAVs). By capitalizing on platform mobility, a virtual array with an effective aperture larger than the number of physical sensors can be synthesized.

Whether implemented with fixed or movable antennas, sensor location errors pose a significant challenge to estimation

performance, with the problem being particularly pronounced in MAs. In these scenarios, the application of conventional subspace-based algorithms, such as MUSIC [10] and ESPRIT [11], leads to degradation in estimation accuracy, as these methods critically depend on the precise knowledge of the array manifold. To solve this problem, two calibration approaches are developed: active calibration [12], [13] and passive calibration [14], [15]. Active calibration utilizes a set of known reference sources to estimate and correct sensor location errors, thereby offering high calibration accuracy. However, in practical scenarios where known reference sources are unavailable, calibration must rely on the received signal information. This self-referential approach is termed passive calibration. In [16], sensor location errors are calibrated using the maximum likelihood (ML) method. More recently, approaches based on the sparse Bayesian learning (SBL) framework are introduced for the joint estimation of DOA and sensor location errors [17], [18], [19], [20].

This paper introduces a signal model for synthesizing a virtual coprime array when there are 2-D sensor location errors in a single sensor movable antenna. Then the virtual coprime array is extended further to a larger virtual ULA. Since the virtual ULA contains holes, the central continuous part is taken as the received signal. Finally, the sensor location errors are calibrated via ML method.

II. SIGNAL MODEL

Under the assumption of ideal conditions, a movable antenna equipped with a single sensor is mounted on an UAV. The UAV is positioned at the nominal location corresponding to the coprime array along the x-axis, while maintaining a stationary position along the y-axis. These positions can be defined as follows:

$$\mathbb{P} = \{(x, 0) \mid x \in \{mNd\}_{m=0}^{M-1} \cup \{nMd\}_{n=0}^{N-1}\}, \quad (1)$$

where $d = \lambda/2$ represents the standard inter-element spacing, with λ denoting the wavelength. M and N are a pair of prime integers satisfying $M < N$. The virtual coprime array consists of $T = M + N - 1$ sensors. Assume K far-field incoherent narrowband sources impinge on the single sensor with angles

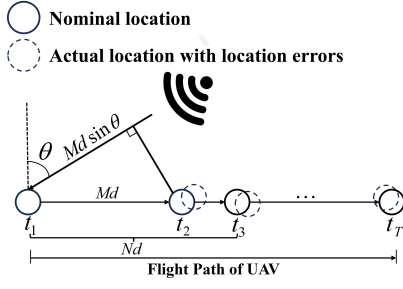


Fig. 1. The flight path of the UAV

of θ_k at time t_p , $k = 1, 2, \dots, K$, $p = 1, 2, \dots, T$. The received signal can be expressed as

$$r_{p,q} = \mathbf{a}(t_p, \theta) \mathbf{s}(q) + z((p-1)Q + q), \quad (2)$$

with:

- $r_{p,q} \in \mathbb{C}^{1 \times 1}$ is the received signal at the p -th time (position) and the q -th pilot; Q represents the number of pilot symbols, $q = 1, 2, \dots, Q$.
- $\mathbf{a}(t_p, \theta) = [a(t_p, \theta_1), a(t_p, \theta_2), \dots, a(t_p, \theta_K)] \in \mathbb{C}^{1 \times K}$ is the directional vector; $a(t_p, \theta_k) = e^{j \frac{2\pi}{\lambda} d_p \sin \theta_k} \in \mathbb{C}^{1 \times 1}$; d_p represents the distance between the p -th position and the initial position.
- $\mathbf{s}(q) \in \mathbb{C}^{K \times 1}$ is the source vector of the q -th pilot symbol emitted by K signal sources;
- $z((p-1)Q + q) \in \mathbb{C}^{1 \times 1}$ is the independent identically distributed additive Gaussian white noise signal with zero mean and variance σ_n^2 received in the $(p-1)Q + q$ -th time slot.

Fig. 1 shows the flight path of the UAV and the locations where data is collected. Therefore, the total received signal at the p -th position can be expressed as

$$\mathbf{r}_p = [r_{p,1}, r_{p,2}, \dots, r_{p,Q}]. \quad (3)$$

The antenna is assumed to remain static at each position during the reception of constant pilot signal, implying the phase rotation due to motion within the sampling interval is negligible. Finally, the received signal of the virtual coprime array can be synthesized as

$$\mathbf{Y} = \begin{bmatrix} \mathbf{r}_1 \\ \mathbf{r}_2 \\ \vdots \\ \mathbf{r}_T \end{bmatrix} = \mathbf{A}(\theta) \mathbf{S} + \mathbf{Z}, \quad (4)$$

with:

- $\mathbf{Y} \in \mathbb{C}^{T \times Q}$ is the received signal matrix;
- $\mathbf{A}(\theta) = [\mathbf{a}(t_1, \theta); \mathbf{a}(t_2, \theta); \dots; \mathbf{a}(t_T, \theta)] \in \mathbb{C}^{T \times K}$ is the directional matrix;
- $\mathbf{S} = [\mathbf{s}(1), \mathbf{s}(2), \dots, \mathbf{s}(Q)] \in \mathbb{C}^{K \times Q}$ is the source matrix;
- $\mathbf{Z} = [\mathbf{z}(1); \mathbf{z}(2); \dots; \mathbf{z}(T)] \in \mathbb{C}^{T \times Q}$ is the noise matrix; $\mathbf{z}(p) = [z((p-1)Q + 1), z((p-1)Q + 2), \dots, z(pQ)] \in \mathbb{C}^{1 \times Q}$

The covariance matrix of the received signal can be expressed as

$$\mathbf{R} = \mathbf{A} \mathbf{R}_s \mathbf{A}^H + \sigma_n^2 \mathbf{I} \approx \frac{1}{Q} \mathbf{Y} \mathbf{Y}^H, \quad (5)$$

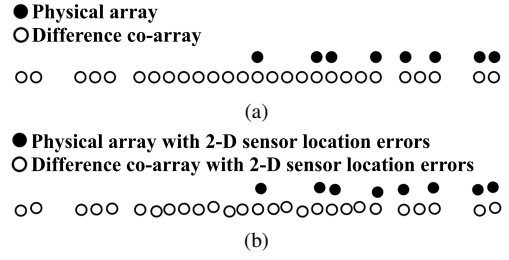


Fig. 2. (a) The nominal locations of the coprime array and the virtual array; (b) The actual locations of the coprime array and the virtual array.

where $\mathbf{R}_s \in \mathbb{C}^{K \times K}$ is the covariance matrix of sources; $\mathbf{I} \in \mathbb{C}^{T \times T}$ is the identity matrix; $(\cdot)^H$ represents the conjugate transpose.

III. 2-D SENSOR LOCATION ERRORS OF VIRTUAL ARRAY

Assume 2-D positional deviation during the flight of the UAV, then the directional matrix can be expressed as

$$\tilde{\mathbf{A}} = [\tilde{\mathbf{a}}(\theta_1), \tilde{\mathbf{a}}(\theta_2), \dots, \tilde{\mathbf{a}}(\theta_k)], \quad (6)$$

where $\tilde{\mathbf{a}}(\theta_k) = [e^{j \frac{2\pi}{\lambda} (d_1 \sin \theta_k + \Delta d_{1,k})}, e^{j \frac{2\pi}{\lambda} (d_2 \sin \theta_k + \Delta d_{2,k})}, \dots, e^{j \frac{2\pi}{\lambda} (d_T \sin \theta_k + \Delta d_{T,k})}]^T$. $\Delta d_{p,k} = \Delta x_p \sin \theta_k + \Delta y_p \cos \theta_k$ is the wave path difference caused by the location error of the p -th position sensor. Δx_p and Δy_p represent the errors of the p -th position relative to the x -axis and y -axis respectively. $(\cdot)^T$ represents the transpose operation. Denote $\mathbf{A}$ and $\mathbf{A}_\Delta$ as the nominal and error directional matrices, respectively. The matrix $\tilde{\mathbf{A}}$ can be written as

$$\tilde{\mathbf{A}} = \mathbf{A} \odot \mathbf{A}_\Delta, \quad (7)$$

where $(\mathbf{A}_\Delta)_{p,k} = e^{j \frac{2\pi}{\lambda} \Delta d_{p,k}}$; $\odot$ denotes Hadamard product. The covariance matrix with sensor location errors can be expressed as

$$\tilde{\mathbf{R}} = (\mathbf{A} \odot \mathbf{A}_\Delta) \mathbf{R}_s (\mathbf{A} \odot \mathbf{A}_\Delta)^H + \sigma_n^2 \mathbf{I}. \quad (8)$$

By vectorizing the matrix $\tilde{\mathbf{R}}$, the received signal of the virtual array can be obtained as

$$\mathbf{v} = \text{vec}(\tilde{\mathbf{R}}) = [\mathbf{A}_v \odot \mathbf{A}_{\Delta v}] \mathbf{w} + \sigma_n^2 \mathbf{i}, \quad (9)$$

where $\mathbf{A}_v = \mathbf{A}^* \boxtimes \mathbf{A}$, $\mathbf{A}_{\Delta v} = \mathbf{A}_\Delta^* \boxtimes \mathbf{A}_\Delta$; $(\cdot)^*$ and $\boxtimes$ represent the complex conjugate and Khatri-Rao product, respectively; $\text{vec}(\cdot)$ denotes matrix vectorization; $\mathbf{w} = [\sigma_1^2, \sigma_2^2, \dots, \sigma_K^2]^T$ is the power of K sources and $\mathbf{i} = \text{vec}(\mathbf{I})$. Let $\tilde{\mathbf{A}}_v = \mathbf{A}_v \odot \mathbf{A}_{\Delta v}$. It is worth noting that the directional matrix $\tilde{\mathbf{A}}_v \in \mathbb{C}^{(T)^2 \times K}$ contains redundancy. By applying averaging and sorting operations to rows with identical position differences, a compacted matrix $\bar{\mathbf{A}}_v \in \mathbb{C}^{1+2M(N-1) \times K}$ is derived, which is equivalent to the directional matrix of the non-continuous ULA with holes [21]. The new virtual sensor position can be expressed as

$$\bar{d}_u = -M(N-1) + u - 1, u = 1, 2, \dots, 2M(N-1) + 1. \quad (10)$$

After the transformation, the new sensor location errors in the exponent of the matrix $\bar{\mathbf{A}}_v$ are approximately equal to a linear combination of the original sensor location errors. The elements in the matrix $\bar{\mathbf{A}}_v$ can be expressed as

$$\bar{\mathbf{A}}_v(u, k) = e^{j \frac{2\pi}{\lambda} (\bar{d}_u \sin \theta_k + \Delta \bar{d}_{u,k})}, \quad (11)$$

$$\begin{aligned}\Delta\bar{d}_{u,k} &= \beta_1\Delta d_{1,k} + \beta_2\Delta d_{2,k} + \dots + \beta_T\Delta d_{T,k} \\ &= \Delta\bar{x}_u \sin \theta_k + \Delta\bar{y}_u \cos \theta_k,\end{aligned}\quad (12)$$

where $\beta_1, \beta_2, \dots, \beta_T$ are a set of numbers that are not all zero. $\Delta\bar{x}_u$ and $\Delta\bar{y}_u$ are respectively a linear combination of Δx_p and Δy_p . Fig. 2 illustrates the nominal difference co-array as well as the difference co-array with location errors. **Lemma 1** provides the support of (12).

Lemma 1: For the exponential function $e^{j\frac{2\pi}{\lambda}\Delta}$, if Δ is much smaller than the wavelength, it can be approximated using the first-order Taylor expansion as follows:

$$e^{j\frac{2\pi}{\lambda}\Delta} \approx 1 + j\frac{2\pi}{\lambda}\Delta. \quad (13)$$

Proof: The original matrix $\tilde{\mathbf{A}}_v$ is expressed as

$$\tilde{\mathbf{A}}_v = [\tilde{\mathbf{a}}_v(\theta_1), \tilde{\mathbf{a}}_v(\theta_2), \dots, \tilde{\mathbf{a}}_v(\theta_K)], \quad (14)$$

$$\tilde{\mathbf{a}}_v(\theta_k) = \left[e^{j\frac{2\pi}{\lambda}(\bar{\Delta}d_{1,1}^k + d_{1,1} \sin \theta_k)}, \dots, e^{j\frac{2\pi}{\lambda}(\bar{\Delta}d_{T,T}^k + d_{T,T} \sin \theta_k)} \right]^T, \quad (15)$$

where $\bar{\Delta}d_{p_1,p_2}^k = \Delta d_{p_1,k} - \Delta d_{p_2,k}$, $d_{p_1,p_2} = d_{p_1} - d_{p_2}$, $p_1, p_2 = 1, 2, \dots, T$. Suppose the u -th row of $\tilde{\mathbf{A}}_v$ is the average of some L rows of $\mathbf{A}_v$, we can obtain

$$\begin{aligned}\bar{\mathbf{A}}_v(u, k) &= \frac{1}{L} \sum_{l=1}^L e^{j\frac{2\pi}{\lambda}(\bar{\Delta}d_{\gamma_l, \eta_l}^k + d_{\gamma_l, \eta_l} \sin \theta_k)} \\ &\approx \frac{1}{L} \sum_{l=1}^L ((1 + j\frac{2\pi}{\lambda}\bar{\Delta}d_{\gamma_l, \eta_l}^k) e^{j\frac{2\pi}{\lambda}d_{\gamma_l, \eta_l} \sin \theta_k}) \\ &= \frac{1}{L} \left(\sum_{l=1}^L e^{j\frac{2\pi}{\lambda}d_{\gamma_l, \eta_l} \sin \theta_k} + \sum_{l=1}^L j\frac{2\pi}{\lambda}\bar{\Delta}d_{\gamma_l, \eta_l}^k e^{j\frac{2\pi}{\lambda}d_{\gamma_l, \eta_l} \sin \theta_k} \right) \\ &= e^{j\frac{2\pi}{\lambda}\bar{d}_u \sin \theta_k} + \frac{1}{L} e^{j\frac{2\pi}{\lambda}\bar{d}_u \sin \theta_k} \sum_{l=1}^L j\frac{2\pi}{\lambda}\bar{\Delta}d_{\gamma_l, \eta_l}^k \\ &= e^{j\frac{2\pi}{\lambda}\bar{d}_u \sin \theta_k} \left[1 + j\frac{2\pi}{L\lambda} \sum_{l=1}^L (\Delta d_{\gamma_l, k} - \Delta d_{\eta_l, k}) \right] \\ &= e^{j\frac{2\pi}{\lambda}\bar{d}_u \sin \theta_k} [1 + j\frac{2\pi}{\lambda}\bar{\Delta}d_{u,k}] \approx e^{j\frac{2\pi}{\lambda}(\bar{d}_u \sin \theta_k + \Delta\bar{d}_{u,k})}\end{aligned}\quad (16)$$

where $\{(\gamma_l, \eta_l), l = 1, 2, \dots, L\}$ represents the set of elements that satisfy the difference of the u -th row. It can be seen that the new sensor location errors are approximately equal to a linear combination of the original sensor location errors, thus completing the proof. $\square$

Therefore, the received signal of the single snapshot virtual ULA can be obtained as

$$\bar{\mathbf{v}} = \bar{\mathbf{A}}_v \mathbf{w} + \sigma_n^2 \bar{\mathbf{I}}, \quad (17)$$

$\bar{\mathbf{v}} \in \mathbb{C}^{2M(N-1)+1 \times 1}$ remains the received signal of a non-continuous virtual ULA. Therefore, it is necessary to take the middle $\bar{T}$ rows of $\bar{\mathbf{v}}$ to obtain the received signal $\mathbf{v}_{nz}$ of the continuous ULA.

IV. 2-D SENSOR LOCATION ERRORS CALIBRATION BASED ON MAXIMUM LIKELIHOOD METHOD

The received signal of the continuous virtual ULA can be written as

$$\mathbf{v}_{nz} = \mathbf{A}_{vv} \mathbf{s}_{vv} + \mathbf{z}_{vv} \quad (18)$$

where $\mathbf{A}_{vv} \in \mathbb{C}^{\bar{T} \times K}$ is the directional matrix of the virtual ULA with $\bar{T}$ sensors and location errors; $\mathbf{s}_{vv} \in \mathbb{C}^{K \times 1}$ is the

source vector; $\mathbf{z}_{vv} \in \mathbb{C}^{\bar{T} \times 1}$ is the noise vector. To estimate the DOA and location errors, the following formula is provided

$$\{\hat{\theta}_k, \{\Delta\hat{\bar{x}}_u, \Delta\hat{\bar{y}}_u\}\} = \arg \min \|\mathbf{v}_{nz} - \mathbf{A}_{vv} \mathbf{s}_{vv}\|^2, \quad (19)$$

where $\Delta\hat{\bar{x}}_u$ and $\Delta\hat{\bar{y}}_u$ represent the location errors at the middle of the ULA, $\bar{u} = 1 + (\bar{T}-1)/2, \dots, 2M(N-1)+1 - (\bar{T}-1)/2$. The ML method consists of two steps. The first step is to estimate the DOA using the position calibrated in the second step. The second step is to calibrate the position using the DOA estimated in the first step.

A. First Step: DOA Estimation

Firstly, the source matrix is estimated using the least square method, which can be expressed as

$$\hat{\mathbf{s}}_{vv} = (\mathbf{A}_{vv}^H \mathbf{A}_{vv})^{-1} \mathbf{A}_{vv}^H \mathbf{v}_{nz}. \quad (20)$$

In the initial estimation, $\mathbf{A}_{vv}^{(0)}$ is constructed as the initial direction matrix, where the angles are obtained through the rough estimation of the received signal $\mathbf{v}_{nz}$ using SS-MUSIC [22], and the sensor location is the nominal location of the ULA. Then, the K sources can be estimated by maximizing the following optimization function

$$\hat{\theta}_k = \arg \max_{\theta} \frac{1}{\bar{T}} \|\mathbf{v}_{nz,k}^H \mathbf{a}_{vv}(\theta)\|^2, \quad (21)$$

where $\mathbf{a}_{vv}(\theta)$ is the steering vector of the ULA with location errors and

$$\mathbf{v}_{nz,k} = \mathbf{v}_{nz} - \mathbf{A}_{vv} \mathbf{s}_{vv,k}, \quad (22)$$

with $\mathbf{s}_{vv,k}$ indicates that the k -th row is set to zero, while the rest of the rows remain unchanged. All DOAs can be obtained by performing K traversal searches on (21). Finally, update $\mathbf{A}_{vv}$. If (19) converges, proceed to the second step; otherwise, repeat the first step.

B. Second Step: Location Calibration

By writing all the elements as summation, (19) can be reconstructed as

$$\{\hat{\theta}_k, \{\Delta\hat{\bar{x}}_u, \Delta\hat{\bar{y}}_u\}\} = \arg \min \sum_{i=1}^{\bar{T}} |\mathbf{v}_{nz}(i) - \mathbf{A}_{vv}(i, :) \mathbf{s}_{vv}|^2. \quad (23)$$

Analyzing i -th virtual sensor, we can obtain:

$$\begin{aligned}& |\mathbf{v}_{nz}(i) - \mathbf{A}_{vv}(i, :) \mathbf{s}_{vv}|^2 \\ &= |\mathbf{v}_{nz}(i) - \sum_{k=1}^K \mathbf{s}_{vv}(k) e^{j\frac{2\pi}{\lambda}[(\bar{x}_i + \Delta\bar{x}_{\bar{u}(i)}) \sin \theta_k + (\bar{y}_i + \Delta\bar{y}_{\bar{u}(i)}) \cos \theta_k]}|^2,\end{aligned}\quad (24)$$

where $\bar{x}_i$ and $\bar{y}_i$ represent the nominal position of the ULA. According to **Lemma 1**, (24) can be further derived as

$$|\bar{v}_{nz} - f \Delta\bar{x}_{\bar{u}(i)} - g \Delta\bar{y}_{\bar{u}(i)}|^2, \quad (25)$$

where,

$$\bar{v}_{nz} = \mathbf{v}_{nz}(i) - \sum_{k=1}^K \bar{\mathbf{A}}_{vv}(i, k) \mathbf{s}_{vv}(k), \quad (26)$$

$$f = j \frac{2\pi}{\lambda} \sum_{k=1}^K \bar{\mathbf{A}}_{vv}(i, k) \mathbf{s}_{vv}(k) \sin \theta_k, \quad (27)$$

$$g = j \frac{2\pi}{\lambda} \sum_{k=1}^K \bar{\mathbf{A}}_{vv}(i, k) \mathbf{s}_{vv}(k) \cos \theta_k, \quad (28)$$

with $\bar{\mathbf{A}}_{vv}$ is nominal directional matrix of the ULA and (25) can be restructured as

$$\arg \min \left\| \bar{\mathbf{v}}_{nz} - \begin{bmatrix} f & g \end{bmatrix} \begin{bmatrix} \Delta \bar{x}_{\bar{u}(i)} \\ \Delta \bar{y}_{\bar{u}(i)} \end{bmatrix} \right\|^2, \quad (29)$$

Obviously, the location errors of the i -th virtual sensor can be expressed as

$$\begin{bmatrix} \Delta \hat{x}_{\bar{u}(i)} \\ \Delta \hat{y}_{\bar{u}(i)} \end{bmatrix} = [\text{Re}\{\boldsymbol{\Omega}^H \boldsymbol{\Omega}\}]^{-1} \text{Re}\{\boldsymbol{\Omega}^H \bar{\mathbf{v}}_{nz}\}. \quad (30)$$

where $\boldsymbol{\Omega} = [f, g]$; $\text{Re}\{\cdot\}$ denotes taking the real part of a complex number. All location errors can be obtained by performing $\bar{T}$ calculations on (24). Finally, update $\mathbf{A}_{vv}$. If (19) converges, proceed to the first step; otherwise, repeat the second step. The DOA estimation and location calibration are summarized in Algorithm 1.

Algorithm 1: 2-D Sensor Location Calibration via ML

Input : $\bar{\mathbf{A}}_{vv}, \mathbf{v}_{nz}$
Output : $\theta, \Delta \hat{x}_{\bar{u}}, \Delta \hat{y}_{\bar{u}}$
while $iter1 < \max_iter1$ **do**
 First Step: DOA Estimation
 while $iter2 < \max_iter2$ **or** $error1 > \varepsilon$ **do**
 Calculate $\hat{\mathbf{S}}_{vv}$ based on (20);
 for $k = k + 1$ **do**
 Angle search based on (21) to obtain $\hat{\theta}_k$;
 Update $\mathbf{A}_{vv}$;
 end
 Update $\rho = \|\mathbf{v}_{nz} - \mathbf{A}_{vv} \hat{\mathbf{S}}_{vv}\|^2$;
 $error1 = |\rho^{(iter2-1)} - \rho^{(iter2)}|$;
 $iter2 = iter2 + 1$.
 end
 Second Step: Location Calibration
 while $iter3 < \max_iter3$ **or** $error2 > \varepsilon$ **do**
 Calculate $\hat{\mathbf{S}}_{vv}$ based on (20);
 for $i = i + 1$ **do**
 Error estimation based on (30) to obtain $\{\Delta \hat{x}_{\bar{u}(i)}, \Delta \hat{y}_{\bar{u}(i)}\}$;
 Update $\mathbf{A}_{vv}$;
 end
 Update $\rho = \|\mathbf{v}_{nz} - \mathbf{A}_{vv} \hat{\mathbf{S}}_{vv}\|^2$;
 $error2 = |\rho^{(iter3-1)} - \rho^{(iter3)}|$;
 $iter3 = iter3 + 1$.
 end
 $iter1 = iter1 + 1$.
end

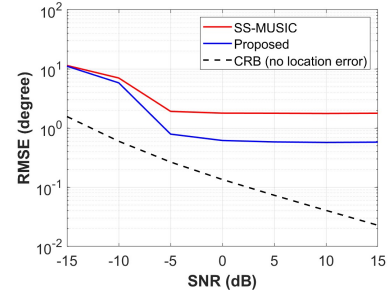


Fig. 3. RMSE of DOA estimation vs. SNR

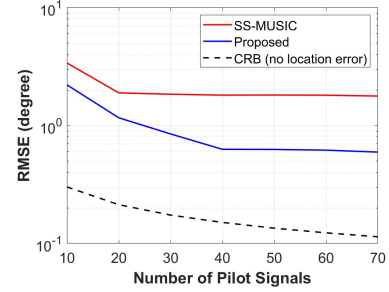


Fig. 4. RMSE of DOA estimation vs. the number of pilot signals

V. SIMULATION RESULTS

We consider that a single sensor movable antenna eventually forms a virtual coprime array with $M = 4$ and $N = 5$. Our algorithm is evaluated with 2 sources, where $\theta_1 = -6^\circ, \theta_2 = 7^\circ$ and The location deviation of the UAV is up to 17% of $\lambda/2$. The Cramér-Rao Bound (CRB) [23] obtained under the condition of no location error is also included as baseline for comparison. The root mean square error (RMSE) is used to evaluate the estimation performance.

With $Q = 50$ considered and SNR ranging from -15 dB to 15 dB. Fig. 3 shows the performance of both algorithms improves with the increase of SNR and converges at high SNR. The accuracy of the proposed algorithm after location calibration is significantly improved compared to SS-MUSIC. Additionally, the proposed algorithm has a significant gap compared to the CRB when there is no location error.

In Fig. 4, the results obtained under fixed SNR of 0 dB are presented for different numbers of pilot signals. As the number of pilot signals increases, the errors of both algorithms decrease and the proposed algorithm consistently outperforms the other one.

VI. CONCLUSION

This paper introduces the synthesis of virtual coprime array with 2-D location errors in a single sensor movable antenna. Subsequently, the virtual coprime array is further transformed into a virtual ULA, and the location errors are calibrated using ML method. The simulation results show that the proposed algorithm can calibrate location errors and improve DOA performance. The problem of the Doppler effect of movable antennas will be studied in future work.

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